

FemPlate: finite-element modal synthesis of plates and membranes of arbitrary shape

Technical documentation

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Abstract

FemPlate is an audio plugin that simulates the linear and weakly nonlinear vibration of a thin plate (or tensioned membrane) of arbitrary user-drawn shape, with mixed boundary conditions. The plate eigenmodes are computed by a non-conforming (Morley) finite-element discretisation and a dense subspace eigensolver, both implemented in the reusable, JUCE-free library `FxmeTools/acoustics`. Sound is produced by a bank of resonant band-pass filters, one per mode, following the modal-synthesis approach of the MechanOdd project. A first-order tension law allows audio-rate retuning of the whole bank without re-solving the eigenproblem, and this same mechanism supports an efficient approximation of geometric (von Kármán–Berger) nonlinearity: amplitude-dependent pitch glide and a cubic inter-mode energy-cascade surrogate, both realtime-safe and unconditionally stable. This document details the mathematical model, the discretisation, the eigensolution, the audio implementation and the nonlinear approximations, with references to the literature.

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1 Continuous model

1.1 Scaled Kirchhoff plate with tension

The transverse deflection $w(\mathbf{x}, t)$ of a thin plate occupying a domain $\Omega \subset \mathbb{R}^2$ is modelled by Kirchhoff–Love theory [1, 2] augmented by a uniform in-plane tension N and two damping mechanisms:

$$\rho h \ddot{w} + c_1 \dot{w} + c_2 \Delta^2 \dot{w} + D \Delta^2 w - N \Delta w = f(\mathbf{x}, t), \quad D = \frac{Eh^3}{12(1 - \nu^2)}, \quad (1)$$

where ρh is the surface density, D the flexural rigidity, ν the Poisson ratio ($\nu = 0.3$ in FemPlate), c_1 a viscous (air/support) damping coefficient and c_2 a stiffness-proportional (material, Kelvin–Voigt-type) damping coefficient. Dividing by ρh and rescaling space and time so that the coefficients of $\ddot{w}$ and $\Delta^2 w$ are unity yields the *scaled* equation actually used throughout the code:

$$\ddot{w} + c_v \dot{w} + c_m \Delta^2 \dot{w} + \Delta^2 w - T \Delta w = f, \quad (2)$$

with a single dimensionless *tension-to-flexural-stiffness* parameter $T \geq 0$. $T = 0$ is a pure plate; $T \rightarrow \infty$ approaches the membrane limit. Frequencies obtained from (2) are dimensionless; the plugin maps the first eigenfrequency onto the user parameter f_1 and all others follow proportionally (§4), so the absolute material constants never need to be specified.

1.2 Boundary conditions

The boundary $\partial\Omega$ is split by the user into segments, each carrying one of four classical conditions:

Name	Essential constraints	Natural conditions	DOFs constrained
Free	—	$M_n = 0, V_n = 0$	—
Simple support	$w = 0$	$M_n = 0$	vertex w
Clamp	$w = 0, \partial_n w = 0$	—	vertex w , edge $\partial_n w$
Sliding support	$\partial_n w = 0$	$V_n = 0$	edge $\partial_n w$

where M_n and V_n are the bending moment and effective (Kirchhoff) shear. Only the *essential* conditions are enforced; the natural ones are satisfied weakly by the variational formulation [3]. The last column anticipates §2.2: the four conditions map one-to-one onto the degrees of freedom of the Morley element, which is the main reason this element was chosen.

2 Finite-element discretisation

2.1 Mesh generation

The library meshes an arbitrary closed polygon (the user’s outline, sampled from a closed Catmull–Rom spline [4] for freehand shapes) with a target element size h :

1. *Boundary resampling.* The outline is walked at arc-length spacing $\approx h$. Vertices where the outline turns by more than 30° are detected as corners and kept exactly, so rectangles remain rectangles. Every boundary sample stores its arc-length parameter $t \in [0, 1)$, which is later used to attach the per-segment boundary conditions.
2. *Interior lattice.* A hexagonal lattice of pitch h fills the bounding box; points closer than $0.62h$ to the boundary or outside the polygon are discarded. A small deterministic jitter ($\pm 0.06h$) breaks the cocircular degeneracies that regular lattices would otherwise present to the triangulator.
3. *Triangulation.* Bowyer–Watson incremental Delaunay insertion [5, 6], with a super-triangle and cavity re-triangulation. The implementation is the plain $O(n^2)$ variant, entirely adequate for the $n \lesssim 2000$ points used here.
4. *Clipping.* Triangles whose centroid falls outside the polygon are removed (this handles non-convex outlines), orientation is normalised to counter-clockwise, and an edge table with adjacency is built. Edges with a single adjacent triangle are boundary edges; they inherit the wrap-aware midpoint of their endpoints’ arc parameters.

2.2 The Morley element

The biharmonic operator requires C^1 continuity for conforming elements, which leads to high-order elements (Argyris, Bell) that are expensive and delicate. FemPlate instead uses the *Morley triangle* [7], the simplest non-conforming plate bending element: on each triangle the deflection is a full quadratic polynomial (6 coefficients) determined by 6 degrees of freedom,

- the deflections w_i at the three vertices,
- the normal derivatives $(\partial_n w)_e$ at the three edge midpoints.

The element is non-conforming (w is discontinuous along edges except at vertices), yet it converges for fourth-order problems, and its eigenvalue approximations converge at rate $O(h^2)$, *from below* [8, 9] — both properties are verified numerically in §2.4. Crucially for FemPlate, its two DOF families are exactly the quantities constrained by the four boundary conditions of §1.2.

Rather than hard-coding the classical closed-form shape functions, each element inverts a 6×6 generalized Vandermonde system: writing the local polynomial in centroid-centred monomials $p(\mathbf{x}) = \sum_m c_m \mu_m(\mathbf{x})$, $\{\mu_m\} = \{1, X, Y, X^2, XY, Y^2\}$, the matrix $C_{im} = \ell_i(\mu_m)$ of the six DOF functionals ℓ_i is inverted by Gaussian elimination with partial pivoting; column j of C^{-1} holds the monomial coefficients of shape function N_j . Centroid-centred coordinates keep C well-conditioned at the element scale. The mid-edge normal is the *global* normal of the edge (stored with $v_0 < v_1$, normal to the left of $v_0 \rightarrow v_1$), so the shared DOF has the same meaning in both adjacent elements.

2.3 Element matrices

With constant curvatures per element (second derivatives of a quadratic), the bending stiffness is exact:

$$K_{ij}^e = A_e \boldsymbol{\kappa}_i^\top \mathbf{D} \boldsymbol{\kappa}_j, \quad \boldsymbol{\kappa}_j = \begin{pmatrix} \partial_{xx} N_j \\ \partial_{yy} N_j \\ 2 \partial_{xy} N_j \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{pmatrix}, \quad (3)$$

(A_e = element area; flexural rigidity is 1 after scaling). The tension (geometric) matrix and the consistent mass matrix are integrated with quadrature rules exact for their integrands:

$$G_{ij}^e = \int_e \nabla N_i \cdot \nabla N_j \, dA \quad \text{3-point midpoint rule (degree 2, exact),} \quad (4)$$

$$M_{ij}^e = \int_e N_i N_j \, dA \quad \text{6-point degree-4 rule [10] (quartic, exact).} \quad (5)$$

G is the discrete counterpart of $-\Delta$ restricted to the plate space and doubles as the *geometric stiffness* familiar from buckling analysis [3]. Global assembly scatters K^e, G^e, M^e into dense symmetric matrices over the *free* DOFs only: essential boundary conditions (§1.2) are applied by elimination, using each boundary vertex's / edge's arc parameter to look up its segment's condition. At a junction between two segments the vertex takes the strongest condition (clamp > simple support > sliding > free).

2.4 Validation

The console test suite (`Tests/FemTests.cpp`) verifies the implementation against classical analytical results [1]:

- simply supported unit square: $\lambda_{mn} = \pi^4(m^2 + n^2)^2$, with the measured $O(h^2)$ convergence from below (error ratio 3.9 when h is halved) and the exact tension coefficient $g_{11} = 2\pi^2$ (§3.2);
- clamped circle: frequency ratios 2.081, 3.414, 3.893 and the absolute fundamental $\omega_{01} = (\beta_{01}/R)^2$, $\beta_{01}^2 = 10.2158$ [1], within 2–5% at moderate mesh density;
- mixed clamped/free boundaries, mode-shape vanishing on the clamped side, and the first-order tension law against an exact re-solve.

3 The eigenvalue problem

3.1 Formulation and solver

Discretisation of (2) (undamped part) yields the symmetric generalized eigenproblem

$$(K + T_0 G) \varphi_k = \lambda_k M \varphi_k, \quad \lambda_k = \omega_k^2, \quad \varphi_k^\top M \varphi_l = \delta_{kl}, \quad (6)$$

solved at a *reference tension* T_0 (the value of the Tension knob when the user presses *Compute*). With n free DOFs ($n \approx \text{vertices} + \text{edges} \approx 4 \times \text{vertices}$, typically 300–3000) and only the lowest $m \leq 128$ modes wanted (capped at $\sim n/6$ so the top modes stay resolved by the mesh), the solver is classical *subspace iteration with Rayleigh–Ritz projection* [11, 3]:

1. Form $A = K + T_0 G$ and the shifted operator $P = A + \sigma M$ with $\sigma = 10^{-5} \text{tr}(A)/\text{tr}(M)$. The shift keeps P positive definite when free (Neumann-type) boundaries leave rigid-body modes in A 's null space. P is Cholesky-factored once ($O(n^3/3)$).
2. Iterate on a block of $p = m + \max(8, m/2)$ vectors: $Z \leftarrow P^{-1} M X$ (one inverse-power step towards the low spectrum), followed by Rayleigh–Ritz on $\text{span}(Z)$: project $A_p = Z^\top A Z$, $M_p = Z^\top M Z$, solve the $p \times p$ generalized problem by Cholesky reduction and cyclic Jacobi [12], and take the Ritz vectors as the next block X .
3. Stop when the wanted eigenvalues are stationary: relative 10^{-6} on the lower half, 10^{-4} on the upper half — the slowest to converge, and the modes whose FEM discretisation error is orders of magnitude larger anyway (both thresholds are far below one cent in frequency); or after 60 iterations.

Implementation notes that matter for speed: the iteration block is stored *transposed* (every iteration vector contiguous in memory, so all $O(n^2p)$ loops stream cache lines), the independent per-vector solves and matrix-vector products are spread over a few worker threads, and the Jacobi solver of the projected $p \times p$ problem uses a *relative* off-diagonal convergence test (the entries scale with $\lambda^2 \sim 10^{12}$; an absolute threshold silently burns the full sweep budget on every call). Together these give a $\sim 4\times$ speed-up over the straightforward implementation (23.6 s $\rightarrow$ 5.6 s for 128 modes on a 931-DOF mesh).

Finally, the computed modal data (eigenvalues, tension coefficients, shapes, statistical tail) is cached inside the plugin state: loading a session or preset republishes the model immediately, with no recomputation. The mesh is not serialised — `generateMesh` is a pure deterministic function of the outline and density, so it is rebuilt on load and checked against the stored vertex count. Eigenvalues below an absolute threshold (10^{-4} ; rigid-body and near-rigid modes of free plates) are dropped. Eigenvectors are mass-normalised, their vertex values are exported for display and for the pointwise evaluations $\varphi_k(\mathbf{x})$ used by the synthesis (linear interpolation on the located triangle; the mid-edge DOFs remain internal). The whole computation runs on a background thread with progress reporting; completed models are published to the audio thread by an atomic pointer swap with generation-based reclamation.

3.2 Audio-rate tension changes without re-solving

The solver additionally returns, for every mode, the *tension sensitivity*

$$g_k = \varphi_k^T G \varphi_k \left(= \int_{\Omega} |\nabla \varphi_k|^2 dA \right), \quad (7)$$

enabling the first-order frequency law used at audio rate:

$$\boxed{\omega_k^2(T) = \lambda_k + (T - T_0) g_k} \quad (8)$$

This is legitimate because T enters the operator *linearly* and the Rayleigh quotient is *stationary* at eigenvectors: perturbing T changes φ_k at first order in $(T - T_0)$, but that eigenvector error affects the eigenvalue only at *second* order. Equation (8) is the standard eigenvalue derivative $\partial \lambda_k / \partial T = \varphi_k^T G \varphi_k$ of structural sensitivity analysis [15], a special case of classical perturbation theory [13, 14]; here it is exact at T_0 with $O((T - T_0)^2)$ error. For the simply supported square the law is exact for all T (the eigenvectors do not depend on T); in general, pressing *Compute* re-anchors T_0 when the knob has travelled far. The validation suite checks (8) against an exact re-solve at $T = 50$ (error $< 1\%$).

Beyond the knob itself, (8) is the enabling mechanism for the nonlinear glide of §5.1: *dynamic* tension is just a fast-moving T .

4 Modal synthesis with resonant filter banks

4.1 Modal decomposition

Expanding $w(\mathbf{x}, t) = \sum_k q_k(t) \varphi_k(\mathbf{x})$ in (2) and using M -orthonormality decouples the dynamics into damped oscillators

$$\ddot{q}_k + 2\zeta_k \omega_k \dot{q}_k + \omega_k^2 q_k = \varphi_k(\mathbf{x}_h) F(t), \quad w(\mathbf{x}_o, t) = \sum_k \varphi_k(\mathbf{x}_o) q_k(t), \quad (9)$$

for a point force F at $\mathbf{x}_h$ and a pickup at $\mathbf{x}_o$ — the standard modal synthesis framework [16, 17, 18, 19]. FemPlate realises each oscillator as a second-order IIR resonator, exactly as in the MechanOdd modal resonators: one constant-peak-gain band-pass biquad per mode (RBJ

“cookbook” coefficients [20]) with $Q_k = 1/(2\zeta_k)$, excited by $\varphi_k(\mathbf{x}_h)F$ and weighted on output by $\varphi_k(\mathbf{x}_o)$. Up to 128 modes are active; the bank is allocation-free and retunable on the audio thread.

4.2 Frequency and damping laws

The user selects the fundamental in hertz and the bank plays at

$$f_k = f_1 \frac{\omega_k(T_{\text{eff}})}{\omega_1(T_{\text{knob}})}, \quad T_{\text{eff}} = T_{\text{knob}} + T_{\text{dyn}}, \quad (10)$$

with all ω_k given by (8). The normalisation reference matters: it is the fundamental at the *static* knob tension, not the instantaneous one. The static Tension knob therefore keeps mode 1 pinned at f_1 and only reshapes the overtone ratios (the timbre morphs from plate towards membrane), whereas the *dynamic* tension of §5.1 shifts the whole spectrum, fundamental included — the audible hardening glide. Normalising by the instantaneous fundamental instead would cancel the glide of mode 1 and leave only the overtone-ratio compression towards the membrane spectrum (g_k/g_1 ratios are lower than the plate ratios $\sqrt{\lambda_k/\lambda_1}$), which is perceived as a *softening* — the opposite of the physical behaviour. So the timbre (inharmonic pattern) is set by geometry, boundary conditions and tension, while f_1 transposes it. Modes falling outside [20 Hz, 0.47 f_s] are muted.

Projecting the two damping terms of (2) onto mode k gives modal damping ratios (using $\Delta^2 \varphi_k \approx \omega_k^2 \varphi_k$ for the material term)

$$\zeta_k = \underbrace{\frac{c_v}{2\omega_k}}_{\text{viscous}} + \underbrace{\frac{c_m \omega_k}{2}}_{\text{material}} = \frac{\zeta_v}{\nu_k} + \zeta_m \nu_k, \quad (11)$$

where the knobs ζ_v, ζ_m are simply the two contributions to the damping ratio of *mode 1*. Viscous damping therefore lets high modes ring relatively longer (constant absolute bandwidth), material damping kills them faster — the two classical, perceptually distinct decay profiles of struck plates [21, 22].

4.3 Excitation, pickup, gain compensation

A mouse hit at $\mathbf{x}_h$ launches a half-sine force pulse (a standard hammer/impact model [21])

$$F(t) = F_0 \sin(\pi t/\tau), \quad 0 \leq t \leq \tau, \quad (12)$$

with duration τ (*Hammer* knob, 0.1–50 ms: short = bright, long = mellow, since the pulse spectrum rolls off above $\approx 1/\tau$) and amplitude F_0 (*Force* knob $\times$ click velocity). Up to 8 hammers overlap polyphonically, each carrying its own weight vector $\varphi_k(\mathbf{x}_h)$. The external audio input is injected identically at the last hit point (*effect mode*), and the output point $\mathbf{x}_o$ is a pair of automatable parameters.

Because a constant-peak band-pass transmits energy proportionally to its bandwidth ($\propto \zeta_k$), lightly damped modes would ring much quieter than damped ones. Each mode is therefore compensated by $\sqrt{\zeta_{\text{ref}}/\zeta_k}$ (clamped to [0.5, 32], $\zeta_{\text{ref}} = 10^{-2}$), the loudness compensation inherited from the MechanOdd modal resonators. Retunes triggered by parameter changes rewrite only the biquad coefficients (the states are preserved), which is inaudible for the small throttled steps used here.

5 Nonlinear approximation

Large-amplitude plate vibration is governed by the von Kármán equations [23, 24]: transverse deflection stretches the mid-plane, producing an amplitude-dependent membrane stress that (i) raises all frequencies (the gong/tom *pitch glide*) and (ii) couples the modes, cascading energy towards high frequencies (the cymbal *crash*, up to wave turbulence [30]). Exact modal von Kármán synthesis requires the cubic coupling tensors $\Gamma_{klmn} \sim O(N^3)$ – $O(N^4)$ storage and work [28, 29] — which is beyond a realtime budget at 128 modes. FemPlate implements two lightweight surrogates, one for each phenomenon.

5.1 Dynamic tension: the Berger approximation

Berger [25] observed that for many plate problems the von Kármán membrane coupling is well approximated by a *spatially uniform* tension proportional to the integrated stretching:

$$\ddot{w} + \Delta^2 w - \underbrace{\left(T_{\text{knob}} + \kappa \int_{\Omega} |\nabla w|^2 dA \right)}_{T_{\text{dyn}}} \Delta w = f. \quad (13)$$

In modal coordinates $\int |\nabla w|^2 = \sum_{kl} q_k q_l \boldsymbol{\varphi}_k^\top G \boldsymbol{\varphi}_l \approx \sum_k g_k q_k^2$: the stretching integral is measured by the very same G and g_k of §3.2, and the resulting uniform T_{dyn} feeds straight into the retune law (8). The Berger model captures the hardening (glide-up-and-relax) behaviour of plates and membranes and is a standard efficient approximation in the nonlinear vibration literature [25, 26]; its perceptual signature on percussion is documented by Legge & Fletcher and Rossing [27, 22].

Amplitude estimation and calibration. The band-pass resonators are output-normalised (constant peak gain), so their internal states are *not* calibrated modal displacements: a naive $\sum_k g_k \langle y_k^2 \rangle$ over filter outputs y_k underestimates the stretching by orders of magnitude (impulse ring amplitudes scale with ζ_k). FemPlate therefore drives the effect from the *audible output* and expresses it as a dimensionless *relative stiffening of mode 1*:

$$\gamma = \min(\gamma_{\text{max}}, \alpha \cdot \text{nonlin} \cdot \langle w^2(\mathbf{x}_o) \rangle), \quad T_{\text{dyn}} = \gamma \frac{\omega_1^2(T_{\text{knob}})}{g_1}, \quad (14)$$

with $\alpha = 8$, $\gamma_{\text{max}} = 4$, and $\langle \cdot \rangle$ a 20-ms one-pole envelope. By construction $\omega_1^2 \mapsto \omega_1^2 (1 + \gamma)$ exactly, and every other mode follows its own physical sensitivity g_k : γ is plate-independent ($\gamma = 0.4$ always means “mode 1 up ≈ 3 semitones”), which makes the knob calibration robust across geometries. Losing the per-mode weighting is a second approximation on top of Berger’s; the qualitative behaviour — glide amplitude tracking the ring energy — is preserved.

Realtime scheduling. γ is updated every 32 samples, slewed by a factor 0.2 per update ($\approx$ ms-scale smoothing), and the biquads are rewritten only when mode 1 would move by more than ≈ 2 cents ($|\Delta\gamma| > 0.0024(1 + \gamma)$, since $\Delta\omega/\omega = \frac{1}{2}\Delta\gamma/(1 + \gamma)$). Coefficient rewrites preserve filter states; the throttling keeps them free of zipper artefacts, following the same strategy as MechanOdd’s gate-morphed damping.

5.2 Mode cascade: a multi-band cubic ladder

A memoryless cubic of a signal band-limited to $[0, B_\omega]$ produces inter-modulation products only up to $3B_\omega$, concentrated near the original band. A single “distort the low modes, inject into the high ones” pass therefore *cannot* reach the top of the bank: with sources below $\sim 12f_1$ the products die out by ~ 3 kHz while the bank extends to 9 kHz, and all that remains audible of

such a scheme is the side effect of its overlap floor — high-frequency damping. The physical cascade climbs the spectrum *iteratively*: products beget products, stage after stage [30].

FemPlate reproduces this with a *windowed* band ladder. The active modes are split into $N_b = 8$ contiguous index bands $\mathcal{B}_0 \dots \mathcal{B}_7$ (equal index ranges are equal frequency ranges, plate modal density being constant); band b is driven by the cubic of the W bands directly below it only (default $W = 4$),

$$w^{(b)}(t) = \sum_{c=b-W}^{b-1} \sum_{k \in \mathcal{B}_c} \varphi_k(\mathbf{x}_o) c_k y_k(t), \quad F_b(t) = A g_b(t) \tanh^3(B w^{(b)}(t)), \quad (15)$$

fed to mode $l \in \mathcal{B}_b$ with weight $\varphi_l(\mathbf{x}_h)/c_l$ (dividing by the bandwidth compensation c_l , which is re-applied on output, makes the audible cascade level independent of the damping settings). The source *window* matters: summing all lower bands would let the loud strike band drive the top bands quasi-directly and the whole spectrum lights up within milliseconds; with the window, each stage extends the spectral reach by at most a factor 3 and energy genuinely climbs rung by rung. The per-band gate $g_b(t) \in [0, 1]$ adds *transfer inertia*: it follows the source presence through an attack/release envelope whose attack time grows with the band’s height on the ladder ($\tau_b = 30 \text{ ms} \times b$ and a 2 s release in the voiced defaults), so the brightening is a progressive glow and the pumping tails off smoothly instead of collapsing with the cube of the decaying ring. Voicing found a hard-saturating transfer most convincing: a large drive ($B = 16$) with a small injection gain ($A = 1.1$, capped at 1.5 — beyond that the level becomes unpleasant). A cubic is the leading nonlinearity of von Kármán dynamics [28]; at low amplitude (15) vanishes as w^3 (the linear model is recovered exactly), and at high amplitude the bands brighten in sequence, reproducing the amplitude-gated crash of cymbals and gongs [22]. All cascade internals (A , B , the source window, the gate times and the overlap floor below) are exposed as plugin parameters while the effect is being voiced; the values quoted here are the defaults. Note also that the material damping law $\zeta \propto \nu$ dominates the high-mode decay when hunting long shimmer: at $\zeta_m = 10^{-4}$ a mode at $\nu = 80$ decays in milliseconds whatever the cascade does.

Modal-overlap floor. The cubic’s inter-modulation products are broadband, but at typical plate dampings the receiving resonances are needles: bandwidth $2\zeta_l f_l$ of a few hertz against a mean mode spacing $\Delta f \approx 0.64 f_1$ (the plate Weyl law gives a constant modal density $dN/d\nu = \pi/2$ for the unit square), i.e. a modal overlap of order 0.1. Most of the injected energy then falls *between* modes and is rejected — heard as distortion of isolated partials rather than a cascade, whereas the crash regime of real cymbals has overlap $\gg 1$ [22]. The target bandwidths are therefore floored, proportionally to the cascade control, at an adjustable fraction (*Overlap*, default 0.1) of the *local* mode spacing (measured on the actual f_k array),

$$\zeta_l \geq \text{cascade} \times \text{Overlap} \times \frac{\frac{1}{2}(f_{l+1} - f_{l-1})}{2f_l}, \quad l \notin \mathcal{B}_0, \quad (16)$$

turning the receiving comb into a quasi-continuum when the cascade is engaged (and, physically consistently, shortening the high modes’ ring — nonlinear energy spreading is itself a loss channel for individual modes). At cascade = 0 the linear model is untouched.

Material-damping compensation. The audible strength of the cascade depends steeply on the material damping: large ζ_m widens the target resonances (overlap floor) and raises their bandwidth compensation, so a knob value that is mild at feather damping saturates the output when hit and pickup are close. Listening tests produced qualitative “maximum usable cascade” limits

ζ_m	$2.4 \cdot 10^{-3}$	10^{-3}	10^{-4}	10^{-5}	$7 \cdot 10^{-6}$
max cascade	0.05	0.1	0.4	0.9	1

which are nearly linear in log-log; the fitted power law

$$s(\zeta_m) = \min\left(1, (2.3 \cdot 10^{-5}/\zeta_m)^{0.64}\right) \quad (17)$$

reproduces the table within $\sim 10\%$. The effective cascade amount is $\text{cascade} \times s(\zeta_m)$ (applied to the injection, the overlap floor and the depletion alike), so the knob keeps its full travel and its maximum tracks the acceptability limit at every damping setting.

Depletion (low-frequency loss). The transfer (15) is additive: it copies energy upward but takes none from the sources, so the struck low modes ring on untouched under the shimmer — audibly two disconnected layers. In the real cascade the low modes *lose* the energy they hand upward. FemPlate models this as an extra dissipation on the source bands: each sample, the filter states of the modes in band b are scaled by

$$d_b = 1 - \text{Deplete} \times \text{cascade} \times \varepsilon_0 \bar{g}_b^\uparrow, \quad \varepsilon_0 = 3 \cdot 10^{-4} \quad (\text{Deplete} = 0.07 \text{ by default}), \quad (18)$$

where $\bar{g}_b^\uparrow$ is the mean gate level of the bands fed by band b : the lows decay exactly when, and as strongly as, they are pumping the highs, gluing the two ends of the spectrum. The mechanism is purely dissipative (it only removes energy), so it cannot affect stability.

Stability by structure. Two earlier designs closed a feedback loop through the full output. Around mode k 's resonance such a loop transmits with gain $|\varphi_k(\mathbf{x}_h)| \cdot |\varphi_k(\mathbf{x}_o)| \cdot c_k$ — and the compensation $c_k \leq 32$ must not be forgotten: an unnormalised loop is super-critical and latches into a full-scale limit cycle the moment the cubic's slope wakes up. Bounding the loop slope by the small-gain theorem [33, 34] restores stability but is so conservative (the bound assumes worst-case coherent feedback at the single worst mode) that the effect becomes inaudible. The ladder (15) dissolves the dilemma structurally: band b receives only from bands $c < b$ and contributes only to bands $c > b$, so the transfer graph over bands is a strict DAG — no loop exists at any stage, stability is unconditional, and the injection gain A is free to be set for audibility ($A = \text{Amp} \times \text{cascade}$, the $\tanh^3$ keeping every stage bounded).

5.3 The statistical mode tail

Even a 128-mode FEM bank remains a sparse comb at the top of the spectrum, and solving for many hundreds of accurate FEM modes is prohibitive (§3) — while a convincing crash needs a dense receiving continuum up to Nyquist. FemPlate therefore fills the bank (to 256 modes in all) with *synthetic* modes above the FEM set, built from asymptotics instead of the discrete operator:

- *Frequencies*: the plate Weyl law gives a constant modal density in ω ; the tail continues at the spacing $\Delta\omega$ measured on the top half of the FEM spectrum, with gaps jittered by $\pm 40\%$ (deterministically seeded) to avoid periodicity artefacts.
- *Tension sensitivities*: for an asymptotic mode of wavenumber κ , $\lambda = \kappa^4$ and $g = \kappa^2$, hence $g = \sqrt{\lambda}$ exactly (verified by the simply supported square, where $g_{mn} = \pi^2(m^2+n^2) = \sqrt{\lambda_{mn}}$) — so the tail obeys the tension law (8) and the Berger glide like any FEM mode.
- *Shapes*: high-frequency eigenfunctions of irregular domains behave like superpositions of random plane waves (Berry's random-wave model [31]); each tail mode carries one, $\varphi(\mathbf{x}) = \sqrt{2/A} \cos(\boldsymbol{\kappa} \cdot \mathbf{x} + \psi)$ with $|\boldsymbol{\kappa}| = \lambda^{1/4}$, random direction and phase, and the $\sqrt{2/A}$ prefactor reproducing the mass normalisation ($\overline{\varphi^2} = 1/A$). Hit and pickup positions therefore still modulate the tail response.

The tail modes enter the synthesis exactly like FEM modes (hammer excitation, damping law, cascade bands, dynamic tension); only the contour display is FEM-only, since tail modes have no mesh shape. The identity-defining low modes stay exact, and the top of the bank becomes the quasi-continuum the cascade ladder needs — the reasoning is that of statistical energy analysis [32]: above a few dozen mode indices, individual eigenfrequencies are perceptually interchangeable and only their density and statistics matter.

6 Implementation summary

Stage	Thread	Notes
Shape / boundary editing	message	ShapeCanvas ; outline + arc-parameterised segments
Mesh generation	message	milliseconds; preview in FemViewComponent
Eigensolution (6)	background	BackgroundTaskRunner ; progress; seconds
Model publication	message → audio	atomic pointer, generation-acknowledged reclamation
Filter bank (9)	audio	allocation-free; retunes throttled
Nonlinearities (14), (15)	audio	decimated γ update; bounded feedback

The numerical core (**FemMesh**, **PlateModes**) is pure C++17 with no JUCE dependency and is validated by a console test suite; it lives in the **FxmeTools** submodule (**FxmeTools/acoustics**) for reuse in other projects.

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